

COMP 102 – Discrete Mathematics

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1. What is Discrete Mathematics?

Discrete Mathematics deals with objects that are **separate and distinct** (countable). It is the foundation for many areas of computer science: algorithms, data structures, databases, cryptography, logic and graph theory.

Although topics can vary, a typical Discrete Mathematics course covers: Set Theory, Relations, Functions, Mathematical Logic, Proof techniques, Counting, Probability, Graph Theory, Trees, Recurrence relations and Boolean Algebra.

Continuous vs Discrete

Continuous	Discrete
Values on a continuous scale	Separate, distinct values
Any two numbers have another in between (e.g. real numbers)	Can be counted (finite or infinite but countable)

Continuous mathematics works with smooth quantities; Discrete mathematics works with separate, countable objects

In continuous mathematics (e.g. calculus) a function can be plotted as a smooth curve. In discrete mathematics we deal with countable sets of points or distinct objects.

2. Introduction to Set Theory

Set Theory was introduced by the German mathematician **Georg Cantor**. It forms the basis of several other fields such as counting theory, relations, graph theory and finite state machines.

Definition of a Set

A set is an unordered collection of distinct objects (called elements or members). The objects can be anything: numbers, letters, people, other sets, etc. Order does not matter, and repeating an element does not change the set.

Sets are usually denoted by capital letters (A, B, S, ...). Elements are listed inside curly braces { }.

3. Ways of Describing / Representing Sets

1. Roster (Tabular) form

List all the elements explicitly, separated by commas, inside curly braces.

$A = \{a, e, i, o, u\}$ (the set of vowels)

$B = \{1, 2, 3, 5, 7, 9\}$ (some odd numbers)

2. Set-builder form

Describe the property that every element of the set must satisfy.

$A = \{x \mid x \text{ is a vowel in the English alphabet}\}$

$B = \{x \mid x \in \mathbb{N}, x \leq 10 \text{ and } x \text{ is odd}\}$

Read "|" or ":" as "such that". So the second example means: the set of all x such that x is a natural number, x is at most 10, and x is odd.

Membership

If an element x belongs to a set S we write $x \in S$. If it does not belong we write $x \notin S$.

Example: If $S = \{1, 2, 3\}$ then $2 \in S$ but $5 \notin S$.

4. Special Sets of Numbers

- $\mathbb{N}$ – Natural numbers (positive integers): $\{1, 2, 3, \dots\}$ (sometimes includes 0)
- $\mathbb{Z}$ – Integers: $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- $\mathbb{Q}$ – Rational numbers (fractions): numbers that can be written as p/q where p, q are integers, $q \neq 0$
- $\mathbb{R}$ – Real numbers (all points on the number line)
- $\mathbb{Z}^+$ or $\mathbb{N}$ – Positive integers

- $\mathbb{Z}^-$ – Negative integers

5. Cardinality of a Set

The **cardinality** of a set S , written $|S|$, is the number of elements in S .

Example: $A = \{1, 2, 3, 4\} \rightarrow |A| = 4$

If a set has infinitely many elements we say its cardinality is infinite and often write $|S| = \infty$.

Equal cardinality / Equivalent sets

Two sets X and Y have the same cardinality (are equivalent) if there exists a one-to-one correspondence (bijection) between their elements. We write $|X| = |Y|$. They may have different elements but the same number of elements.

Example: $A = \{1, 2, 3\}$ and $B = \{a, b, c\} \rightarrow |A| = |B| = 3$ (equivalent sets)

6. Important Types of Sets

Finite set

A set that contains a definite (countable and limited) number of elements. Example: $S = \{x \in \mathbb{N} \mid x \leq 50\}$ is finite.

Infinite set

A set that contains infinitely many elements. Example: the set of all natural numbers $\mathbb{N}$, or $S = \{x \in \mathbb{N} \mid x > 10\}$.

Singleton (Unit) set

A set that contains exactly one element. Example: $S = \{5\}$ or $S = \{x \in \mathbb{N} \mid 7 \leq x \leq 7\}$.

Empty (Null) set

A set that contains no elements at all. It is denoted by $\emptyset$ or $\{\}$. Its cardinality is zero: $|\emptyset| = 0$. Example: the set of all natural numbers less than 0 is empty.

Universal set

The set of all objects under consideration in a particular discussion or application. It is usually denoted by U . Every other set in that context is a subset of U . Example: if we are talking about animals on Earth, U could be the set of all animals.

Equal sets

Two sets are equal if they contain exactly the same elements (order does not matter). Example: $A = \{1, 2, 3\}$ and $B = \{3, 1, 2\} \rightarrow A = B$.

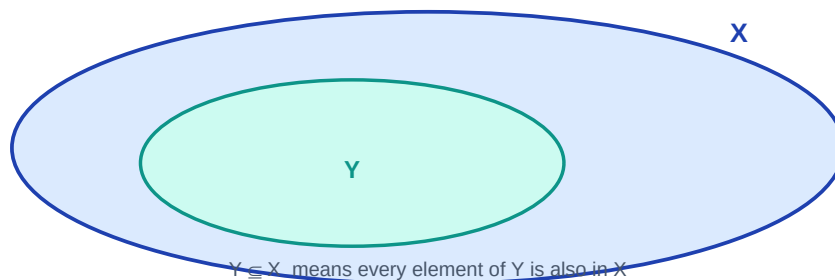
Equivalent sets

Two sets are equivalent if they have the same cardinality (same number of elements), even if the elements themselves are different. Example: $\{1, 2, 3\}$ and $\{a, b, c\}$ are equivalent but not equal.

7. Subsets and Proper Subsets

Subset

Set Y is a subset of set X (written $Y \subseteq X$) if every element of Y is also an element of X . X is then called a superset of Y .



Example: Let $X = \{1, 2, 3, 4, 5, 6\}$ and $Y = \{1, 2, 3\}$. Every element of Y is in X , so $Y \subseteq X$.

Proper subset

Y is a proper subset of X (often written $Y \subset X$) if $Y \subseteq X$ and $Y \neq X$. That is, X has at least one element that is not in Y .

In the example above Y is a proper subset of X because X contains 4, 5 and 6 which are not in Y .

Notes: Every set is a subset of itself ($A \subseteq A$). The empty set is a subset of every set ($\emptyset \subseteq A$).

8. Quick Revision Checklist

- A set is an unordered collection of distinct objects.
- Roster form lists elements; Set-builder form describes a property.
- $x \in S$ means x is an element of S ; $x \notin S$ means it is not.

- $|S|$ = number of elements in S (cardinality).
- Finite / Infinite / Singleton / Empty / Universal are important types of sets.
- $Y \subseteq X$ means every element of Y is also in X .
- Equal sets have exactly the same elements; Equivalent sets have the same number of elements.

These notes expand the Set Theory portion of COMP 102 (Silas Momanyi). Master the definitions and the difference between $\subseteq$, $\subset$, equal and equivalent sets — they appear frequently in CATs and exams.